INCH-POUND

MIL-M-16149G(NU)
12 March 1990
SUPERSEDING
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17 August 1981

MILITARY SPECIFICATION

*MITTEN, WELDERS

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for one type of welders mitten with a one finger leather gauntlet.

1.2 Classification. The mitten shall be procured in one size only, for either the left or right hand as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
* NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction 2 Way and 4 Way (Partial)
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
DDD-L-20	- Label; for Clothing, Equipage, and Tentage (General Use)

MILITARY

* MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
* MIL-STD-147	- Palletized Unit Loads
* MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US POSTAL MANUAL

(Copies are available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606-5808.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Leather. The leather shall be flesh split cattlehide, conforming to Type IV, class 2 or Type VIII of KK-L-2004 suitable for use in gloves. Elongation requirements shall not apply. The color shall be the color resulting from chrome tanning.

3.3.1.1 Thickness. The thickness of the leather shall be not less than 3 1/2 ounces nor more than 5 ounces, except for the fourchette and welt strips of the mitten where a minimum of 3 ounces is permitted. Thickness determinations shall be conducted as specified in 4.4.2.

3.3.2 Thread. The thread for seaming and stitching the mitten shall be cotton, conforming to type IA1, 4 ply, ticket numbers 16 and 20, of V-T-276.

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3.3.3 Gauntlet lining. The gauntlet lining shall consist of an unbleached cotton duck fabric conforming to the requirements for type I, number 8, hard textured duck of CCC-C-419; or a gauntlet lining fabric composed of two thicknesses of unbleached cotton fabric securely laminated together. The laminated fabric shall be composed of a cotton twill fabric weighing not less than 3.7 ounces per square yard and a plain weave cotton backing fabric weighing not less than 1.7 ounces per square yard. The finished fabric (laminated) shall contain not less than 14 ounces per square yard of natural rubber, synthetic rubber, or synthetic resin, and weigh not less than 19.4 ounces per square yard (see 4.4.3).

3.3.4 Hand liner. The inside back of the mitten shall have a knitted fleece fabric liner. The knitted fabric shall have an unbleached cotton ground with a napped wool pile face, light gray in color, and shall weigh not less than 6.0 ounces per square yard when tested as specified in 4.4.3.

* 3.4 Label. Each mitten shall be marked with a combination identification and size label conforming to type III or IV, class 4, of DDD-L-20, except that the size is not applicable. The color of the marking medium shall be dark blue or black and the markings shall show fastness to accelerated laundering only. The size of the lettering shall be 1/8 (± 1/16) inch in height, and shall bear the following inscription:

MITTEN, WELDERS
CONTRACT NO: DLA-000-000-0000 (Example)
STOCK NO: 8415-00-000-0000 (Example)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (IF OTHER THAN CONTRACTOR)

3.5 Design. The mitten shall be of a one-finger slip-on style, consisting of a one-piece back, welted inseams, gauntlet, fabric cuff lining, and an insulated lining on inside back of mitten extending from the wrist to the tip of the fingers. The mitten shall also have a fourchette between the index finger and the three-finger section. (See Figure 1)

3.5.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.6 Dies and patterns. The dies or patterns for cutting all parts of the mitten shall be furnished by the contractor, and shall be of the proper proportions to provide for a good fitting mitten conforming to the measurements specified in Table II.

3.7 Construction

3.7.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation, any one may be used.

3.7.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

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3.7.3 Type 301 stitching. Ends of stitching shall be backstitched or backtacked not less than 3/8 inch, except where ends are caught in other seams and stitching, or when ends of the stitching can overlap. Stitching shall overlap not less than 3/8 inch.

3.7.3.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/8 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped stitches or run-offs noted during inspection of the item (in-process or end item) shall be repaired. The stitching shall start a minimum of 3/8 inch back of the defective area, continue over the defective area and continue a minimum of 3/8 inch beyond the defective area onto the existing stitching. Loose or tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.7.4 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of all seams or stitching, when not caught in other seams or stitching, shall have a 1/2 to 3/4 inch chain extending beyond each end. All repairs shall be in accordance with 3.7.3.1a and 3.7.3.1.b.

3.8 Manufacturing operations requirements. The mitten shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations, provided that the finished mitten is identical to that produced by following the exact sequence as listed in Table I. Minor modifications are permitted where necessary when using automatic equipments. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

3.9 Abbreviations in table of operations. The abbreviations used in Table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double
chnstch	-	chainstitch

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TABLE I - CONSTRUCTION OF MITTEN

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	Cutting a. The leather components of the mitten shall be block or clicker cut (see 3.3.1). Before cutting, the bends shall be dampened and all the stretch in the lengthwise direction pulled out. Each leather component shall be cut in the length direction and shall consist of the following: 1. Palm, face of thumb, and first (index) finger (1 piece of leather). 2. Face of thumb reinforcement (1 piece of leather). 3. Back of thumb (1 piece of leather). 4. Gauntlet-palm side (1 piece of leather). 5. Entire back of mitten including back of first (index) finger and gauntlet (1 piece of leather). 6. Fourchette (1 piece of leather). 7. Leather welt for insertion into thumb seam (1 piece of leather). 8. Leather welt for insertion into fourchette seam (1 piece of leather). 9. Leather welt for insertion into gauntlet to palm seam (1 piece of leather). 10. Leather welt for insertion into closing seam (1 piece of leather). b. Cut the two gauntlet lining pieces to the sufficient length and width, to conform to operations 6 and 7 (see 3.3.3). c. Cut the knitted liner to the sufficient length and width, to conform to operation 7 (see 3.3.4). d. The leather welt pieces shall be 3/16 to 1/4 inch wide.				
2.	Replacement of defective components During the cutting and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly matched components.				

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TABLE I - CONSTRUCTION OF MITTEN

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
3.	Attach fourchette Stitch one side of fourchette to back leather piece (first (index) finger to remainder of mitten) 1/8 to 3/16 inch from the raw edges, catching a leather welt strip in the seam.	301 or 401	SSj-1	8-10	16/4 16/4	16/4 20/4
4.	Attach thumb reinforcement piece a. Superimpose the thumb reinforcement piece over the face of thumb, the top edge and palm edges even.					
	b. Stitch across the bottom edge of the reinforcement piece, 1/8 (± 1/16) inch from the raw edge.	301 or 401	LSa-1	8-10	16/4 16/4	16/4 20/4
5.	Join thumb to palm Stitch back of thumb piece to face of thumb with reinforcement piece 1/8 to 3/16 inch from raw edges, catching a leather welt strip in one continuous operation, wrist edge to wrist edge.	301 or 401	SSj-1	8-10	16/4 16/4	16/4 20/4
6.	Join gauntlet to palm a. Superimpose the straight edge of gauntlet lining (see 3.3.3) over the leather gauntlet piece (palm-side) with the wrist edge and side edges even. The lining shall extend to within 3/8 (± 1/16) inch from top edge of gauntlet opening.					
	b. Stitch the assembled gauntlet to palm at wrist edge 1/8 to 3/16 inch from raw edges, catching a leather welt strip and the bottom edge of thumb in seam.	301 or 401	SSj-1	8-10	16/4 16/4	16/4 20/4
	c. Turn in the top edge of leather gauntlet piece at the opening 1/2 (± 1/8) inch, and stitch 1/8 inch from the raw edge through all plies.	301	SSbc-1	8-10	16/4	16/4
7.	Assemble back of mitten a. Stitch the bottom edge of back gauntlet lining (see 3.3.3) to the wrist edge of knitted liner (see 3.3.4), 1/16 to 1/8 inch from the raw edge.	301 or 401	SSa-1	8-10	16/4 16/4	16/4 20/4

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TABLE I - CONSTRUCTION OF MITTEN

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	b. Position the knitted liner evenly around the edges of the first (index) finger and the three finger sections, with the gauntlet lining to knitted liner seam facing the inside back of the mitten. The curved portion of the gauntlet lining shall extend to within 3/8 (± 1/8) inch from gauntlet opening.					
	c. Turn in the top edge of leather gauntlet at the opening 1/2 (± 1/8) inch, and stitch to the lining 1/8 inch from the raw edge.	301	EFa-1	8-10	16/4	16/4
NOTE:	Operation 9 may be performed in lieu of 6c and 7c.					
8.	Closing Join the front to back of mitten, and stitch all around the side of mitten from top edge to top edge of gauntlet opening 1/8 to 3/16 inch from raw edges, catching the opposite edge of the fourchettes, the knitted liner, the gauntlet lining and a leather welt strip in the seam.	301 or 401	SSj-1	8-10	16/4 16/4	16/4 20/4
9.	Hem gauntlet (Alternate operation of 6c and 7c) Turn in the top edge of leather gauntlet piece at the opening 1/2 (± 1/8) inch and stitch 1/8 inch from raw edge through all plies.	301	SSbc-1	8-10	16/4	16/4
10.	Marking (see 3.4) Each mitten shall be marked on the gauntlet lining (back-side), with the printing centered and positioned 1 1/4 (± 1/8) inches from the last line to the top edge of gauntlet opening.					
11.	Finishing a. Turn the glove to its finished position. The leather welt strips shall not protrude more than 1/8 inch from any finished seam.					
	b. The mitten shall be neatly laid-off on a heated form.					
12.	Trimming and cleaning Trim thread ends, except for the required 1/2 to 3/4 inch chain extending beyond the end of type 401 stitching (see 3.7.4), and remove all loose thread. Remove all spots and stains.					

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3.10 Measurements. The finished mitten shall conform to the measurements as shown in Table II. All measurements and tolerances are expressed in inches.

Table II - Finished measurements

Measurement	Dimension	Tolerance
Width of palm (A)	5 1/2	± 1/4
Width of gauntlet cuff opening (B)	7 1/2	± 1/4
Length of thumb (C)	3 3/8	± 1/4
Length of first finger (D)	5 1/2	± 1/4
Length of palm (E)	8 1/2	± 1/4
Overall length (F)	15 1/4	± 1/4

NOTE: A through F refer to Figure 1.

A. Measurement taken across palm, approximately 1 inch above thumb crotch, folded edge to folded edge.
B. Measurement taken across gauntlet cuff opening, folded edge to folded edge.
C. Measurement taken from thumb crotch to seam tip of thumb.
D. Measurement taken from tip of first (index) finger to thumb crotch.
E. Measurement taken from tip of mitten to gauntlet palm seam.
F. Measurement taken from tip of mitten to top edge of gauntlet cuff.

3.11 Workmanship. The finished mitten shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects listed in 4.4.4.1 and 4.4.4.2.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are performed as specified. The government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.4.2 Intermediate testing. The cut leather parts of the mitten shall be tested for thickness before fabrication (see Table I, operation 1). Any cut part that is not within the specified thickness range indicated in 3.3.1.1, shall constitute a defect. Thickness of the leather shall be determined in accordance with method 1011 of FED-STD-311. The sample unit shall be one cut part each. The lot size shall be expressed in terms of one cut part. The inspection level shall be S-3 and the acceptable quality level (AQL) shall be 4.0 test failures per hundred units.

4.4.3 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, components and materials listed in Table III shall be tested for the characteristics specified and in accordance with the referenced test methods of FED-STD-191. Testing for all the requirements listed in Table III is applicable to the sample unit. For the gauntlet laminated fabric, the lot size shall be expressed in units of yards and the sample unit shall be 1/2 yard, full width. For the knitted liner fabric, the lot size shall be expressed in units of pounds and the sample unit shall be 1/2 yard, full width. The sample size shall be as follows:

Lot size	Sample units
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

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Table III - Component testing

Component	Characteristics	Requirement paragraph	Test method
Gauntlet laminated fabric	Material identification of face and backing fabric	3.3.3	1200 1/
	Weight (oz per sq yd) of face and backing fabric	3.3.3	5041 1/
	Weave on face and backing fabric	3.3.3	Visual 2/
	Finished fabric, weight of rubber	3.3.3	5041 1/
	Finished fabric, total weight (oz. per sq. yd)	3.3.3	5041
Knitted liner fabric	Material identification	3.3.4	1100
	Finished weight	3.3.4	5041

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be performed per sample unit, and the results reported as "pass or "fail".

4.4.4 Examination of the end item. Examination of the end item shall be in accordance with 4.4.4.1 and 4.4.4.2.

4.4.4.1 Visual examination.

Examine	Defect	Classification Major	Classification Minor
Material (Leather)	Not split leather	X	
	Any hole (except pinhole or needle hole), cut, tear or rip	X	
	Brittle or bony area; thin or weak area;	X	
	-Seriously affecting serviceability	X	
	-Affecting serviceability, but not seriously		X
	Grub or tick marks, cuts		
	-Seriously affecting serviceability	X	
	-Affecting serviceability, but not seriously		X
	Loose, spongy, or flanky:		
	-Seriously affecting serviceability	X	
	-Affecting serviceability, but not seriously		X
	Three or more needle holes or pinholes		X
	Wrinkles:		
	-Seriously affecting serviceability	X	
	-Affecting serviceability, but not seriously		X
	Open vein	X	

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Examine	Defect	Classification Major	Minor
(Gauntlet Lining)	Not type specified	X	
	Any hole, cut, tear or rip	X	
	Any float, smash, loose end(s) or slub		X
	Any broken or missing pick or end		X
	Any mend in fabric (except restitching of seams)		X
	Laminated cloth not securely laminated	X	
(Knitted Hand Liner)	Not type or color specified	X	
	Any hole, cut, tear or run	X	
Cleanness	Any appreciable spot or stain		X
Components and Assembly	Any component part or operation omitted	X	
	Any component not as specified	X	
	Any component part misplaced or not serving intended purpose		X
	Any component part not securely affixed, e.g., gauntlet loosely attached to palm of mitten		X
	Any workmanship damage, e.g., scissor or knife cut	X	
	Needle chew(s)	X	
Seams and Stitching	Any open seam, except on gauntlet hem	X	
	Open stitching on gauntlet hem		X
	NOTE: The seam or stitching shall be classified open when one or more stitches joining a seam or a line of stitching are broken or when two or more continuous skipped or runoff stitches occur.		
	Repairs of open seams not as specified		X
	Not specified seam or stitch type	X	
	Any line of stitching omitted	X	
	Any line of stitching misplaced or not beginning or ending where specified	X	
	Any part of mitten caught in an unrelated operation	X	
	Loose tension resulting in a loose seam	X	
	Tight tension (stitches break or cutting of leather results when normal strain is applied in the direction of the seam or stitching)	X	

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Examine	Defect	Classification Major	Minor
Seams and Stitching (cont'd)	End of 301 stitching not backstitched or overlapped when not caught in another row of stitching, or back-stitched or overlapped less than 3/8 inch		X
	End of seam or stitching produced with stitch type 401, when not caught in other seam or stitching:		
	1. Chain extends less than 1/2 inch or more than 3/4 inch beyond each end		X
	2. Chain does not extend beyond each end	X	
	Stitches per inch (to be scored only when the condition exists on the major portion of seam or stitching):		
	1. One or two stitches per inch less than the minimum		X
	2. Three or more stitches per inch less than the minimum	X	
	3. More than the specified maximum number of stitches per inch, damaging leather	X	
	4. More than the specified maximum number of stitches per inch, but not damaging leather		X
	Gage of stitching irregular or not as specified:		
	1. Seriously affecting serviceability	X	
	2. Affecting serviceability, but not seriously		X
Assembly Detail	Any mitten not neatly laid-off		X
	Thread ends not trimmed throughout the mitten (except where required to extend beyond end of type 401 stitching)		X
	Any thumb, first (index) finger or remaining finger areas distorted or twisted, affecting comfort of wearer	X	
	Any welt protruding more than 1/8 inch from seam		X
	Any welt omitted	X	
	Any welt misplaced or not serving intended purpose		X
	Not constructed with specified number of pieces of leather, knitted liner, and gauntlet linings	X	
Marking	Omitted, incorrect, illegible, or incomplete	X	
	Misplaced	X	X

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4.4.4.2 Dimensional examination. The mitten shall be examined for dimensional defects. Any dimension that is not within the established tolerance, shall be classified as a defect (see 3.10).

4.4.4.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the corresponding acceptable quality levels (AQL's), expressed in defects per 100 units for visual and dimensional examination, shall be as follows. The lot size shall be expressed in units of one mitten. The sample unit for this examination shall be one mitten of each hand procured.

	Inspection level	AQL's Major	AQL's Total
For defects applicable to 4.4.4.1	II	2.5	6.5
For defects applicable to 4.4.4.2	S-2	-	4.0

4.4.5 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2)

5.1.1 Level A. Six (6) mittens for one hand only, shall be evenly stacked in a bundle with the palms facing up. Each bundle shall be securely tied at each end with cotton tape or twine.

5.1.2 Level C. Mittens shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twenty-four (24) mittens for one hand only, packaged as specified in 5.1, shall be packed within a fiberboard shipping container assembled, closed, and reinforced conforming to type CF, class weather-resistant, variety DW, grade V2s, size 1A, of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic, of MIL-B-17757. Level A packages shall be packed on edge, with each bundle reversed one in length, four in width, and one in depth within the shipping container.

5.2.2 Level B. Twenty-four (24) mittens for one hand only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety DW, grade 200, size 1A, of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic, of MIL-B-17757. Level A packages shall be packed on edge, with each bundle reversed one in length, four in width, and one in depth within the shipping container.

5.2.3 Level C. Item packaged, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall marked in accordance with MIL-STD-129.

* 5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding methods C and D or Film Bonding F or G. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

* 5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items are intended for wear by the Department of Defense personnel.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Whether mitten is required for left hand or right hand (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Levels of preservation and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples. For access to samples, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

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* 6.5 Subject term (key word) listing.

Gauntlet, one-finger
Handwear, leather
Handwear, protective
Handwear, work

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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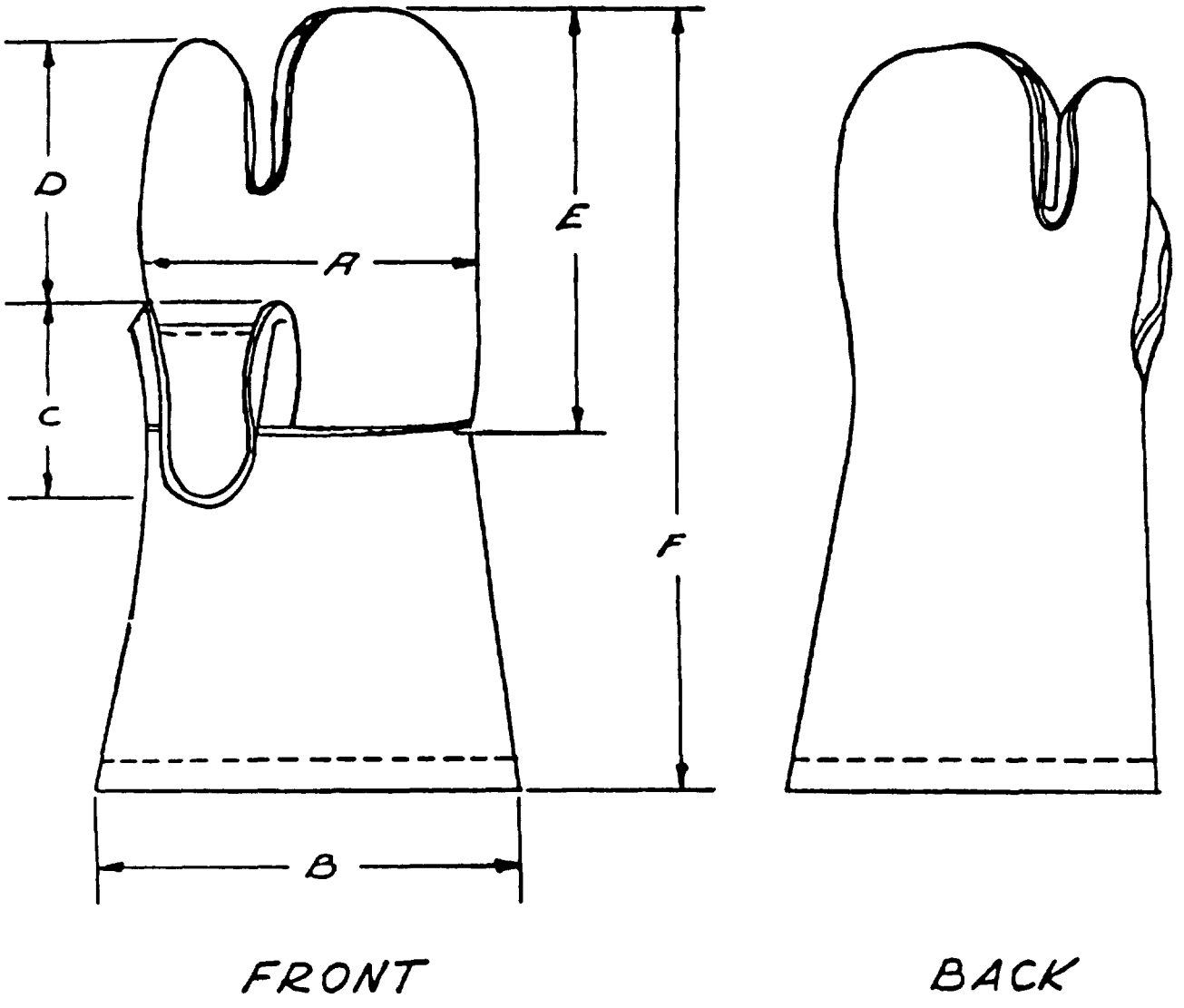


FIGURE I - MITTEN, WELDERS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-16149G(NU)	MITTEN, WELDERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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